Faxonius rusticus
(Rusty Crayfish)
Crustaceans-Crayfish
Native Transplant



Tennessee Wildlife Resources Agency ©

Faxonius rusticus (Girard, 1852)

Common name: Rusty Crayfish

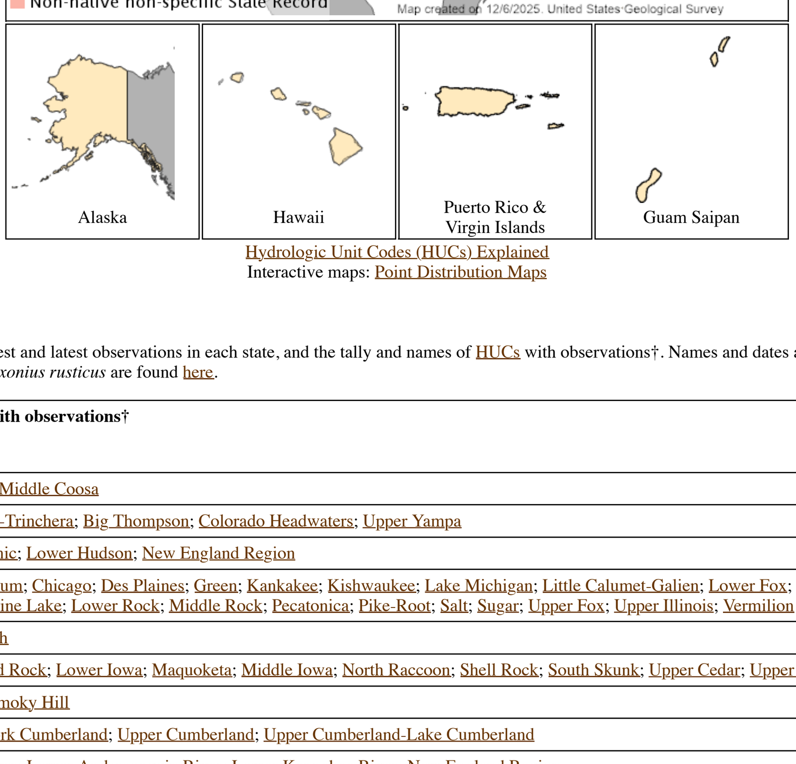
Synonyms and Other Names: *Orconectes rusticus* (Girard, 1852). *Faxonius rusticus* underwent a reclassification in August 2017, changing the genus of non-cave dwelling *Orconectes* to *Faxonius* (Crandall and De Grave 2017).

Taxonomy: available through ITIS

Identification: Brownish-green body with dark, rusty-red spots on either side of carapace (Page 1985). Dark brown section on dorsal abdomen (Gunderson 2008). Large chelae with an oval gap when closed. The dactyl is smooth and S-shaped (Gunderson 2008). Tips of chelae are red with black bands (Page 1985).

Size: Reaches a maximum of 10 cm in length, with males tending to be larger than females. Reaches maturity at about 3.5 cm (Gunderson 2008).

Native Range: Ohio River basin, spanning tributaries in Western Ohio, Indiana, Kentucky, and Northern Tennessee; cryptogenic in Lake Erie (Creaser 1931, Hobbs 1974, Momot et al. 1978, Page 1985, Hobbs et al. 1989, Taylor 2000).



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Faxonius rusticus* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AL	2000	2000	2	Cahaba; Middle Coosa
CO	2009	2023	4	Alamosa-Trinchera; Big Thompson; Colorado Headwaters; Upper Yampa
CT	1989	2016	3	Housatonic; Lower Hudson; New England Region
IL	1906	2024	20	Apple-Plum; Chicago; Des Plaines; Green; Kankakee; Kishwaukee; Lake Michigan; Little Calumet-Galien; Lower Fox; Lower Illinois-Lake Chautauqua; Lower Illinois-Senachwine Lake; Lower Rock; Middle Rock; Pecatonica; Pike-Root; Salt; Sugar; Upper Fox; Upper Illinois; Vermilion
IN	1985	1995	1	St. Joseph
IA	1977	2024	9	Lake Red Rock; Lower Iowa; Maquoketa; Middle Iowa; North Raccoon; Shell Rock; South Skunk; Upper Cedar; Upper Wapsipinicon
KS	2021	2021	1	Lower Smoky Hill
KY	1982	2015	3	South Fork Cumberland; Upper Cumberland; Upper Cumberland-Lake Cumberland
ME	1989	2013	4	Dead River; Lower Androscoggin River; Lower Kennebec River; New England Region
MD	2003	2011	4	Conococheague-Opequon; Lower Susquehanna; Mid Atlantic Region; Monocacy
MA	1989	1989	1	New England Region
MI	1990	2020	37	Au Gres-Rifle; Au Sable; Betsie-Platte; Black-Macatawa; Black-Presque Isle; Boardman-Charlevoix; Brule; Carp-Pine; Cass; Cheboygan; Dead-Kelsey; Detroit; Huron; Lake Erie; Lake Huron; Lake Michigan; Lake St. Clair; Lake Superior; Lone Lake-Ocqueoc; Lower Grand; Manistee; Menominee; Muskegon; Ontonagon; Ottawa-Stony; Pere Marquette-White; Pine; Raisin; Saginaw; St. Clair; St. Joseph; St. Marys; Tacoosh-Whitefish; Tahquamenon; Thunder Bay; Tittabawassee; Upper Wisconsin
MN	1967	2025	33	Baptism-Brule; Big Fork; Cannon; Clearwater-Elk; Cloquet; Crow; Crow Wing; Des Moines Headwaters; East Fork Des Moines; Elk-Nokasippi; Kettle; La Crosse-Pine; Lake Superior; Leech Lake; Little Fork; Lower Rainy; Lower St. Croix; Mississippi Headwaters; Otter Tail; Platte-Spunk; Prairie-Willow; Rainy Headwaters; Red Lakes; Root; Rum; Rush-Vermillion; Snake; St. Louis; Twin Cities; Upper Cedar; Upper St. Croix; Vermilion; Zumbro
NE	2007	2016	2	Big Papillion-Mosquito; Lewis and Clark Lake
NV	2016	2016	1	Las Vegas Wash
NH	1989	1989	1	New England
NJ	1968	2019	5	Hackensack-Passaic; Mid-Atlantic Region; Middle Delaware-Mongaup-Brodhead; Middle Delaware-Musconetcong; Rondout
NM	1989	1989	*	
NY	1977	2024	19	Chemung; Chenango; East Branch Delaware; Grass; Hackensack-Passaic; Housatonic; Hudson-Wappinger; Lake Champlain; Lower Hudson; Middle Hudson; Mohawk; Oneida; Oswego; Rondout; Schoharie; Seneca; Upper Delaware; Upper Hudson; Upper Susquehanna
NC	1999	2015	3	Upper Broad; Upper Catawba; Upper Little Tennessee
OH	1897	2010	8	Ashtabula-Chagrin; Cedar-Portage; Hocking; Huron-Vermilion; Lake Erie; Licking; Sandusky; Upper Ohio
OR	2005	2020	3	Lower John Day; Upper John Day; Upper Malheur
PA	1976	2024	24	Bald Eagle; Conococheague-Opequon; Crosswicks-Neshaminy; Lackawaxen; Lake Erie; Lehigh; Lower Juniata; Lower Susquehanna; Lower Susquehanna-Penns; Lower Susquehanna-Swatara; Lower West Branch Susquehanna; Middle Delaware-Musconetcong; Monocacy; Owego-Wappasening; Pine; Raystown; Schuylkill; Shenango; Sinnemahoning; Tioga; Upper Delaware; Upper Juniata; Upper Susquehanna-Lackawanna; Upper Susquehanna-Tunkhannock
SD	2015	2015	1	Lewis and Clark Lake
TN	1989	2006	2	Emory; Powell
VT	1989	2025	4	Lake Champlain; Lamoille River; Waits River-Connecticut River; White River
WV	1977	2011	3	Lower Kanawha; Raccoon-Symmes; Twelvepole
WI	1957	2021	43	Bad-Montreal; Baraboo; Beartrap-Nemadji; Black; Black-Presque Isle; Brule; Buffalo-Whitewater; Castle Rock; Coon-Yellow; Des Plaines; Door-Kewaunee; Duck-Pensaukee; Eau Claire; Flambeau; Jump; La Crosse-Pine; Lake Dubay; Lake Michigan; Lake Winnebago; Lower Chippewa; Lower Fox; Lower St. Croix; Lower Wisconsin; Manitowoc-Sheboygan; Menominee; Middle Rock; Milwaukee; Namekagon; Oconto; Ontonagon; Pecatonica; Peshtigo; Pike-Root; Red Cedar; South Fork Flambeau; Sugar; Upper Chippewa; Upper Fox; Upper Fox; Upper Rock; Upper St. Croix; Upper Wisconsin; Wolf
WY	2003	2021	3	Glendo Reservoir; Lower Laramie; Upper Cheyenne

Table last updated 12/8/2025

† Populations may not be currently present.

* HUCs are not listed for states where the observation(s) cannot be approximated to a HUC (e.g. state centroids or Canadian provinces).

Ecology: *Faxonius rusticus* inhabits lakes, ponds, and streams, preferring areas with rocks, logs, or other debris for shelter. Clay, silt, sand, gravel, and rock all serve as suitable bottom types. However, *F. rusticus* prefers cobble habitat, which allows it to hide if necessary (Taylor and Redmer 1996). This species can thrive in areas of high flow or in standing water, but unlike other species of crayfish that can burrow in the sediment when water conditions decline, the rusty crayfish must have clear, well-oxygenated water year-round to survive (Capelli 1982 and Gunderson 2008). It is usually found at water depths < 1 meter, though it has been found as deep as 14.6 meters in Lake Michigan (Taylor and Redmer 1996). Adults typically occupy pool areas of >20 cm depth, while juveniles are usually found in shallower areas (<15 cm depth) bordering stream edges (Butler and Stein 1985).

Mature rusty crayfish mate in late summer, early fall, or early spring. The female stores sperm transferred from one or more males until its eggs are ready to be fertilized—usually by late spring when water temperatures begin to increase (Berrill and Arsenault 1984). Therefore, it is possible for a single mature female carrying viable sperm to begin a new population if she is released into a suitable habitat. Rusty crayfish females can lay between 80 and 575 eggs (Gunderson 2008). Eggs hatch in three to six weeks depending on water temperature. Juveniles stay with the female for several weeks after hatching (Berrill 1978) and reach full maturity the following year upon completion of about eight to ten molt cycles. After maturity is reached, growth slows greatly, with males typically molting twice per year and females molting once. In the spring, the male molts into a sexually inactive from (Form II) and returns to its sexually active form (Form I) in the summer (Gunderson 2008). The expected lifespan of *F. rusticus* is 3-4 years.

In its native range within the Ohio River valley, *F. rusticus* may seasonally be exposed to water temperatures ranging from close to 0°C up to 39°C; however, it prefers water temperatures between 20 and 25°C (Mundahl and Benton 1990). The maximum growth rate of juveniles is thought to occur at water temperatures between 26 and 28°C, while the maximum juvenile survival rate occurs at temperatures between 20 and 22°C. Therefore, adults will often displace juveniles into warmer habitats to favor maximum growth rate as a means of improving fecundity and competitive abilities (Mundahl and Benton 1990). At temperatures greater than 30°C, *F. rusticus* has been observed digging burrows in the sand beneath rocks near shore as a means of escaping the heat (Mundahl 1989).

Faxonius rusticus individuals feed as shredders, scrapers, collectors, and predators (Lorman and Magnuson 1978). This species is an opportunistic consumer of a variety of aquatic plants, benthic invertebrates, detritus (decaying plants and animals, including associated bacteria), periphyton (algae and microbes attached to objects submersed in water), fish eggs, and small fish (Lorman 1980). Juveniles tend to feed on benthic invertebrates, such as mayflies, stoneflies, midges, and side-swimmers, more often than do adults (Hanson et al. 1990, Momot 1992). Among the options of invertebrate prey for adults, snails are a primary target (Lodge and Lorman 1987).

Means of Introduction: Human activity best explains the presence of the rusty crayfish in areas outside of its native range. Angler bait bucket emptying is thought to be the primary cause of introduction and species spread (Berrill 1978, Crocker 1979, Butler and Stein 1985, Lodge et al. 1986, Hobbs et at. 1989, Lodge et al. 1994, Kerr et al. 2005; Kilian et al. 2012). The rusty crayfish is also commonly sold to schools and biological supply houses, leading to the potential for uninformed release into the wild (Gunderson 2008; Larson and Olden 2008; Kilian et al. 2012). Intentional release into water bodies by commercial crayfish harvesters is another suspected cause of its range expansion (Wilson et al. 2004). A further mechanism of human facilitated introduction is the intentional establishment of this species in lakes as a means of removing nuisance weeds (Magnuson et al. 1975). Once introduced to a new body of water, this species can move an average of 29 meters per day (Byron and Wilson 2001) and colonize the entire littoral zone up to 12 meters depth (Wilson et al. 2004).

Status: *Faxonius rusticus* is established in twenty states: Colorado (Illinois Natural History Survey 2011), Connecticut (Mills et al. 1997), Iowa (Leon et al. 2016); Illinois (Michigan State University 2015); Maryland (Maryland Department of Natural Resources 2012; Kilian 2013); Maine (Hobbs 1989; sighting reports); Michigan (Michigan State University 2015); Minnesota (Passe 2014); North Carolina (Fullerton and Watson 2001; North Carolina Wildlife Resources Commission 2017); eastern Nebraska (M. Wright pers. comm.); southern Nevada (sighting reports); northern New Jersey (Walker 2002); New York (Walker 2002; Dresser et al. 2016); Ohio (Peters 2010); Oregon (Sorenson et al. 2012); Pennsylvania (iMapInvasives 2016); South Dakota (South Dakota Game, Fish and Parks 2015); Vermont (Caduto 2011); Wisconsin (Wisconsin Department of Natural Resources 2015); and West Virginia (Jezerinac et al. 1994; Loughman 2012).

Its status is unknown in Massachusetts, New Hampshire, and Tennessee, as the only reported introductions are from Hobbs (1989).

Extirpated in Wyoming (Wyoming Game and Fish Department 2015).

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological | **Economic** | **Other**

Faxonius rusticus can out-compete and displace native crayfish and reduce resource availability (Capelli 1982; Page 1985; Taylor and Redmer 1996; Bobeldyk and Lamberti 2008). *Faxonius rusticus* has the potential to impact periled species, such as the endangered Queensnake (Reid and Nocera 2015). In laboratory experiments Welch (2014) found that *F. rusticus* increases turbidity. Dresser et al. (2016) found that native crayfish will spend more time exhibiting agnostic behaviors in the presence of *F. rusticus*.

During mesocosm experiments, *F. rusticus* reduced the abundance of the native snail species *Physa gyrina* and *Lymnaea stagnalis;* when *F. rusticus* co-occured with *Cipangopaludina chinensis* (the Chinese mysterysnail), *Lymnaea stagnalis* was extirpated from the mesocosm (Johnson et al., 2009).

Remarks: Found in streams, lakes, and ponds with varying substrates from silt to rock and plenty of debris for cover; needs permanent water, they generally do not burrow to escape dry periods. Breeding occurs in the fall and eggs laid the following spring, hatching within several weeks. The introduction of one female carrying viable sperm could start a new population. Reisinger et al. (2017) found that juveniles from the nonindigenous range have greater plasticity in behavior than juveniles from the native range, resulting in more active juveniles in the nonindigenous range. *Faxonius rusticus* reduces macroinvertebrate density in streams, but does not alter community composition (Kuhlmann 2016). Potential for *F. rusticus* to spread through estuaries as it has been shown to survive at salinities of 15ppt (Bazer et al. 2016). However, a barrier to dispersal may be water velocity, as *F. rusticus* does not perform well at stream velocities of 66 cm sec-1 (Perry and Jones 2018).

Faxonius rusticus underwent a reclassification in August 2017, changing the genus of non-cave dwelling *Orconectes* to *Faxonius* (Crandall and De Grave 2017).

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Other Resources:
Global Invasive Species Database

Minnesota Sea Grant

Michigan DNR-Life History

Minnesota DNR

Wisconsin DNR

Maryland DNR

Indiana DNR

ANSRP

NatureServe

Ontario Identification Guide

Animal Diversity Web

Great Lakes Waterlife

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